

Docker Commands Guide for DevOps Engineers

1. Verify Docker Installation

Check Docker version:

```
docker --version
```

Detailed information:

```
docker version
```

Docker system information:

```
docker info
```

2. Working with Images

List all images:

```
docker images
```

Pull an image from Docker Hub:

```
docker pull nginx
```

Pull a specific version:

```
docker pull nginx:latest
```

Remove an image:

```
docker rmi nginx
```

Remove multiple images:

```
docker rmi image1 image2
```

Remove all images:

```
docker rmi $(docker images -q)
```

Search images:

```
docker search ubuntu
```

3. Working with Containers

Run a container:

```
docker run nginx
```

Run container in detached mode:

```
docker run -d nginx
```

Run container with custom name:

```
docker run -d --name my-nginx nginx
```

Port mapping:

```
docker run -d -p 8080:80 nginx
```

Interactive container:

```
docker run -it ubuntu bash
```

List running containers:

```
docker ps
```

List all containers:

```
docker ps -a
```

Stop container:

```
docker stop container_id
```

Start container:

```
docker start container_id
```

Restart container:

```
docker restart container_id
```

Kill container:

```
docker kill container_id
```

Delete container:

```
docker rm container_id
```

Force delete:

```
docker rm -f container_id
```

Delete all stopped containers:

```
docker container prune
```

4. Container Inspection

Container details:

```
docker inspect container_id
```

Container logs:

```
docker logs container_id
```

Follow logs:

```
docker logs -f container_id
```

Last 100 log lines:

```
docker logs --tail 100 container_id
```

Execute command inside container:

```
docker exec -it container_id bash
```

Container processes:

```
docker top container_id
```

Container resource usage:

```
docker stats
```

5. Building Images

Build image:

```
docker build -t myapp .
```

Build with version tag:

```
docker build -t myapp:v1 .
```

List build history:

```
docker history image_name
```

6. Docker Volumes

Create volume:

```
docker volume create myvolume
```

List volumes:

```
docker volume ls
```

Inspect volume:

```
docker volume inspect myvolume
```

Remove volume:

```
docker volume rm myvolume
```

Mount volume:

```
docker run -d  
-v myvolume:/data  
nginx
```

7. Bind Mounts

Mount local directory:

```
docker run -d
-v /home/user/data:/app/data
nginx
```

8. Docker Networks

List networks:

```
docker network ls
```

Create network:

```
docker network create mynetwork
```

Inspect network:

```
docker network inspect mynetwork
```

Connect container:

```
docker network connect mynetwork container_name
```

Remove network:

```
docker network rm mynetwork
```

9. Docker Compose

Start services:

```
docker-compose up
```

Detached mode:

```
docker-compose up -d
```

Rebuild containers:

```
docker-compose up --build
```

Stop services:

```
docker-compose down
```

View services:

```
docker-compose ps
```

View logs:

```
docker-compose logs
```

Follow logs:

```
docker-compose logs -f
```

10. Save and Load Images

Save image:

```
docker save myapp > myapp.tar
```

Load image:

```
docker load < myapp.tar
```

Export container:

```
docker export container_id > container.tar
```

Import container:

```
docker import container.tar myimage
```

11. Docker Cleanup

Remove unused containers:

```
docker container prune
```

Remove unused images:

```
docker image prune
```

Remove unused volumes:

```
docker volume prune
```

Remove everything unused:

```
docker system prune -a
```

12. Docker Registry

Login to Docker Hub:


```
docker login
```

Tag image:

```
docker tag myapp:v1 username/myapp:v1
```

Push image:

```
docker push username/myapp:v1
```

Pull image:

```
docker pull username/myapp:v1
```

Logout:

```
docker logout
```

13. Monitoring Commands

Docker events:

```
docker events
```

Docker resource usage:

```
docker stats
```

Docker disk usage:

```
docker system df
```

14. Most Common DevOps Commands

Check containers:

```
docker ps -a
```

Check images:

```
docker images
```

Check logs:

```
docker logs -f container_name
```

Enter container:

```
docker exec -it container_name bash
```

Build image:

```
docker build -t app:v1 .
```

Run container:

```
docker run -d -p 8080:80 app:v1
```

Push image:

```
docker push username/app:v1
```

Cleanup:

```
docker system prune -a
```

DevOps Interview Commands

```
docker ps
docker ps -a
docker images
docker logs
docker exec
docker inspect
docker stats
docker build
docker run
docker stop
docker rm
docker network ls
docker volume ls
docker-compose up
docker-compose down
docker system prune
```

These are the commands most frequently used by DevOps Engineers in CI/CD pipelines, Kubernetes environments, container troubleshooting, and production support.